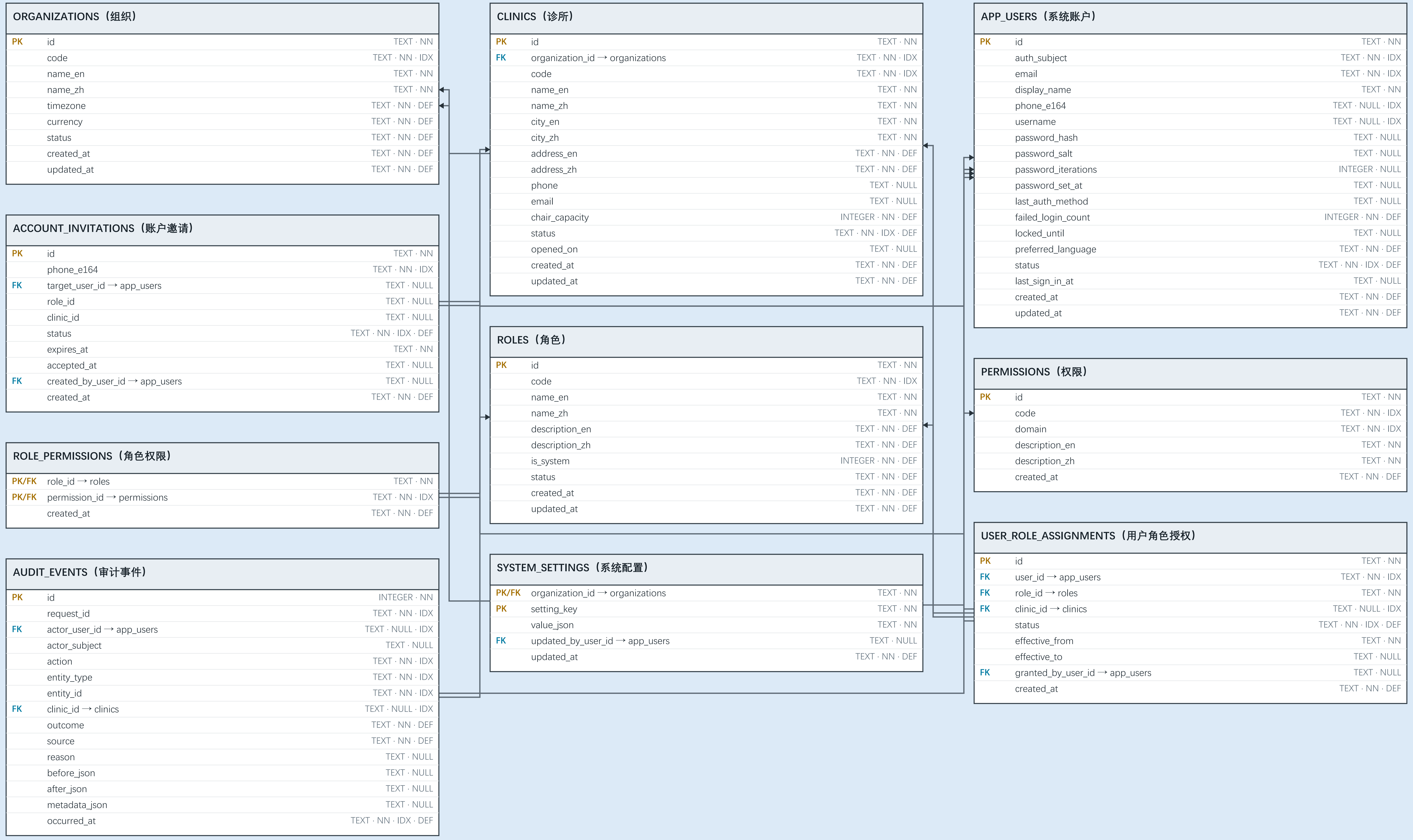
ORGANIZATIONS（组织）
PK id TEXT · NN
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
timezone TEXT · NN · DEF
currency TEXT · NN · DEF
status TEXT · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
ACCOUNT_INVITATIONS（账户邀请）
PK id TEXT · NN
phone_e164 TEXT · NN · IDX
FK target_user_id → app_users TEXT · NULL
role_id TEXT · NULL
clinic_id TEXT · NULL
status TEXT · NN · IDX · DEF
expires_at TEXT · NN
accepted_at TEXT · NULL
FK created_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
ROLE_PERMISSIONS（角色权限）
PK/FK role_id → roles TEXT · NN
PK/FK permission_id → permissions TEXT · NN · IDX
created_at TEXT · NN · DEF
AUDIT_EVENTS（审计事件）
PK id INTEGER · NN
request_id TEXT · NN · IDX
FK actor_user_id → app_users TEXT · NULL · IDX
actor_subject TEXT · NULL
action TEXT · NN · IDX
entity_type TEXT · NN · IDX
entity_id TEXT · NN · IDX
FK clinic_id → clinics TEXT · NULL · IDX
outcome TEXT · NN · DEF
source TEXT · NN · DEF
reason TEXT · NULL
before_json TEXT · NULL
after_json TEXT · NULL
metadata_json TEXT · NULL
occurred_at TEXT · NN · IDX · DEF
CLINICS（诊所）
PK id TEXT · NN
FK organization_id → organizations TEXT · NN · IDX
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
city_en TEXT · NN
city_zh TEXT · NN
address_en TEXT · NN · DEF
address_zh TEXT · NN · DEF
phone TEXT · NULL
email TEXT · NULL
chair_capacity INTEGER · NN · DEF
status TEXT · NN · IDX · DEF
opened_on TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
ROLES（角色）
PK id TEXT · NN
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
description_en TEXT · NN · DEF
description_zh TEXT · NN · DEF
is_system INTEGER · NN · DEF
status TEXT · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
SYSTEM_SETTINGS（系统配置）
PK/FK organization_id → organizations TEXT · NN
PK setting_key TEXT · NN
value_json TEXT · NN
FK updated_by_user_id → app_users TEXT · NULL
updated_at TEXT · NN · DEF
APP_USERS（系统账户）
PK id TEXT · NN
auth_subject TEXT · NN · IDX
email TEXT · NN · IDX
display_name TEXT · NN
phone_e164 TEXT · NULL · IDX
username TEXT · NULL · IDX
password_hash TEXT · NULL
password_salt TEXT · NULL
password_iterations INTEGER · NULL
password_set_at TEXT · NULL
last_auth_method TEXT · NULL
failed_login_count INTEGER · NN · DEF
locked_until TEXT · NULL
preferred_language TEXT · NN · DEF
status TEXT · NN · IDX · DEF
last_sign_in_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PERMISSIONS（权限）
PK id TEXT · NN
code TEXT · NN · IDX
domain TEXT · NN · IDX
description_en TEXT · NN
description_zh TEXT · NN
created_at TEXT · NN · DEF
USER_ROLE_ASSIGNMENTS（用户角色授权）
PK id TEXT · NN
FK user_id → app_users TEXT · NN · IDX
FK role_id → roles TEXT · NN
FK clinic_id → clinics TEXT · NULL · IDX
status TEXT · NN · IDX · DEF
effective_from TEXT · NN
effective_to TEXT · NULL
FK granted_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
AUTH_IDENTITIES（外部身份）
PK id TEXT · NN
FK user_id → app_users TEXT · NN · IDX
provider TEXT · NN · IDX
provider_subject TEXT · NN · IDX
profile_json TEXT · NN · DEF
last_used_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
AUTH_OAUTH_STATES（OAuth 状态）
PK id TEXT · NN
provider TEXT · NN
state_hash TEXT · NN · IDX
return_to TEXT · NN · DEF
expires_at TEXT · NN · IDX
consumed_at TEXT · NULL
created_at TEXT · NN · DEF
AUTH_OTP_CHALLENGES（验证码挑战）
PK id TEXT · NN
phone_e164 TEXT · NN · IDX
purpose TEXT · NN · IDX
code_hash TEXT · NN
code_salt TEXT · NN
expires_at TEXT · NN · IDX
consumed_at TEXT · NULL
attempt_count INTEGER · NN · DEF
max_attempts INTEGER · NN · DEF
request_ip_hash TEXT · NULL
created_at TEXT · NN · IDX · DEF
NOTIFICATION_EVENTS（通知事件）
PK id TEXT · NN
FK appointment_id → appointments TEXT · NULL · IDX
channel TEXT · NN
template_code TEXT · NN
recipient_masked TEXT · NN
locale TEXT · NN · DEF
status TEXT · NN · DEF
idempotency_key TEXT · NN · IDX
queued_at TEXT · NN · IDX
sent_at TEXT · NULL
delivered_at TEXT · NULL
failure_code TEXT · NULL
created_at TEXT · NN · DEF
AUTH_SESSIONS（认证会话）
PK id TEXT · NN
FK user_id → app_users TEXT · NN · IDX
token_hash TEXT · NN · IDX
auth_method TEXT · NN
expires_at TEXT · NN · IDX
revoked_at TEXT · NULL
last_seen_at TEXT · NULL
ip_hash TEXT · NULL
user_agent_hash TEXT · NULL
created_at TEXT · NN · DEF
PATIENT_ACCOUNT_LINKS（患者账户绑定）
PK id TEXT · NN
FK user_id → app_users TEXT · NN · IDX
FK patient_id → patients TEXT · NN · IDX
verification_method TEXT · NN
verified_at TEXT · NN
created_at TEXT · NN · DEF
PATIENTS（患者）
PK id TEXT · NN
patient_code TEXT · NN · IDX
name_en TEXT · NN · IDX
name_zh TEXT · NN · IDX
date_of_birth TEXT · NN
gender TEXT · NN
phone TEXT · NN · IDX
email TEXT · NULL
government_id_last4 TEXT · NULL
preferred_language TEXT · NN · DEF
communication_channel TEXT · NN · DEF
FK primary_clinic_id → clinics TEXT · NULL · IDX
status TEXT · NN · DEF
last_visit_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PATIENT_TIMELINE_EVENTS（患者时间线事件）
PK id TEXT · NN
FK patient_id → patients TEXT · NN · IDX
FK clinic_id → clinics TEXT · NULL
event_type TEXT · NN
occurred_at TEXT · NN · IDX
title_en TEXT · NN
title_zh TEXT · NN
description_en TEXT · NN · DEF
description_zh TEXT · NN · DEF
reference_type TEXT · NULL · IDX
reference_id TEXT · NULL · IDX
status TEXT · NN
amount_fen INTEGER · NULL
metadata_json TEXT · NN · DEF
created_at TEXT · NN · DEF
EMERGENCY_CONTACTS（紧急联系人）
PK id TEXT · NN
FK patient_id → patients TEXT · NN · IDX
contact_name TEXT · NN
relationship_en TEXT · NN
relationship_zh TEXT · NN
phone TEXT · NN
priority INTEGER · NN · IDX · DEF
is_authorized_contact INTEGER · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PATIENT_CONSENTS（患者同意记录）
PK id TEXT · NN
FK patient_id → patients TEXT · NN · IDX
consent_type TEXT · NN · IDX
status TEXT · NN · IDX
policy_version TEXT · NN · IDX
captured_at TEXT · NN
expires_at TEXT · NULL
source TEXT · NN · DEF
created_at TEXT · NN · DEF
PATIENT_ALLERGIES（患者过敏记录）
PK id TEXT · NN
FK patient_id → patients TEXT · NN · IDX
substance_en TEXT · NN
substance_zh TEXT · NN
reaction_en TEXT · NN
reaction_zh TEXT · NN
severity TEXT · NN · IDX
status TEXT · NN · IDX · DEF
verified_at TEXT · NULL
FK verified_by_dentist_id → dentists TEXT · NULL
notes_en TEXT · NN · DEF
notes_zh TEXT · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PATIENT_DUPLICATE_REVIEWS（重复患者复核）
PK id TEXT · NN
FK source_patient_id → patients TEXT · NN · IDX
FK candidate_patient_id → patients TEXT · NN · IDX
match_score INTEGER · NN · IDX
match_reasons_json TEXT · NN
status TEXT · NN · IDX · DEF
FK reviewed_by_user_id → app_users TEXT · NULL
reviewed_at TEXT · NULL
resolution_notes TEXT · NULL
created_at TEXT · NN · DEF
牙医、资质与治疗目录
DENTISTS（牙医）
PK id TEXT · NN
employee_code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
title_en TEXT · NN
title_zh TEXT · NN
email TEXT · NN · IDX
phone TEXT · NULL
hire_date TEXT · NN
employment_status TEXT · NN · IDX · DEF
languages_json TEXT · NN · DEF
biography_en TEXT · NN · DEF
biography_zh TEXT · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
TREATMENT_QUALIFICATION_REQUIREMENTS（治疗资质要求）
PK/FK treatment_id → treatment_catalog TEXT · NN
PK/FK qualification_type_id → qualification_typ··· TEXT · NN · IDX
requirement_level TEXT · NN · DEF
created_at TEXT · NN · DEF
QUALIFICATION_TYPES（资质类型）
PK id TEXT · NN
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
category TEXT · NN
description_en TEXT · NN · DEF
description_zh TEXT · NN · DEF
status TEXT · NN · DEF
created_at TEXT · NN · DEF
DENTIST_CLINIC_ASSIGNMENTS（牙医诊所任职）
PK id TEXT · NN
FK dentist_id → dentists TEXT · NN · IDX
FK clinic_id → clinics TEXT · NN · IDX
is_primary INTEGER · NN · DEF
effective_from TEXT · NN
effective_to TEXT · NULL
weekly_hours INTEGER · NN · DEF
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
DENTIST_QUALIFICATIONS（牙医资质证书）
PK id TEXT · NN
FK dentist_id → dentists TEXT · NN · IDX
FK qualification_type_id → qualification_typ··· TEXT · NN
issuing_body TEXT · NN
reference_number TEXT · NN · IDX
issued_on TEXT · NN
expires_on TEXT · NULL · IDX
verification_status TEXT · NN · DEF
verified_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
TREATMENT_CATALOG（治疗项目目录）
PK id TEXT · NN
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
category TEXT · NN · IDX
description_en TEXT · NN · DEF
description_zh TEXT · NN · DEF
duration_minutes INTEGER · NN
standard_price_fen INTEGER · NN
estimated_cost_fen INTEGER · NN
version INTEGER · NN · IDX · DEF
effective_from TEXT · NN
effective_to TEXT · NULL
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
排班、资源与预约
CLINIC_CHAIRS（诊所牙椅）
PK id TEXT · NN
FK clinic_id → clinics TEXT · NN · IDX
code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
room_name_en TEXT · NN
room_name_zh TEXT · NN
equipment_json TEXT · NN · DEF
is_accessible INTEGER · NN · DEF
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
APPOINTMENTS（预约）
PK id TEXT · NN
appointment_code TEXT · NN · IDX
FK patient_id → patients TEXT · NN · IDX
FK clinic_id → clinics TEXT · NN · IDX
FK dentist_id → dentists TEXT · NN · IDX
FK chair_id → clinic_chairs TEXT · NULL · IDX
FK treatment_id → treatment_catalog TEXT · NN
scheduled_start_at TEXT · NN · IDX
scheduled_end_at TEXT · NN · IDX
duration_minutes INTEGER · NN
status TEXT · NN · IDX · DEF
booking_channel TEXT · NN · DEF
priority TEXT · NN · DEF
confirmation_state TEXT · NN · DEF
risk_flag TEXT · NN · DEF
notes TEXT · NN · DEF
cancellation_reason_code TEXT · NULL
appointment_version INTEGER · NN · DEF
FK created_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
REQUESTED_TREATMENT_ITEMS（预约治疗诉求）
PK id TEXT · NN
FK plan_id → treatment_plans TEXT · NN · IDX
FK visit_id → clinical_visits TEXT · NN · IDX
FK treatment_id → treatment_catalog TEXT · NN
sequence INTEGER · NN · IDX
tooth_code TEXT · NULL
surfaces_json TEXT · NN · DEF
units INTEGER · NN · DEF
priority TEXT · NN · DEF
rationale_en TEXT · NN · DEF
rationale_zh TEXT · NN · DEF
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
CLINIC_OPERATING_HOURS（诊所营业时间）
PK/FK clinic_id → clinics TEXT · NN
PK day_of_week INTEGER · NN · IDX
opens_at TEXT · NN
closes_at TEXT · NN
slot_minutes INTEGER · NN · DEF
is_closed INTEGER · NN · IDX · DEF
updated_at TEXT · NN · DEF
DENTIST_AVAILABILITY_RULES（牙医常规可用规则）
PK id TEXT · NN
FK dentist_id → dentists TEXT · NN · IDX
FK clinic_id → clinics TEXT · NN · IDX
day_of_week INTEGER · NN · IDX
starts_at TEXT · NN
ends_at TEXT · NN
effective_from TEXT · NN
effective_to TEXT · NULL
recurrence_rule TEXT · NN · DEF
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
APPOINTMENT_STATUS_HISTORY（预约状态历史）
PK id TEXT · NN
FK appointment_id → appointments TEXT · NN · IDX
from_status TEXT · NULL
to_status TEXT · NN
reason_code TEXT · NULL
reason_text TEXT · NULL
FK actor_user_id → app_users TEXT · NULL
occurred_at TEXT · NN · IDX
metadata_json TEXT · NN · DEF
created_at TEXT · NN · DEF
WAITLIST_OFFERS（候补邀约）
PK id TEXT · NN
FK waitlist_entry_id → waitlist_entries TEXT · NN · IDX
FK released_appointment_id → appointments TEXT · NN · IDX
FK clinic_id → clinics TEXT · NN
FK dentist_id → dentists TEXT · NN
FK chair_id → clinic_chairs TEXT · NULL
offered_start_at TEXT · NN
offered_end_at TEXT · NN
compatibility_score INTEGER · NN
reasons_json TEXT · NN
status TEXT · NN · IDX · DEF
expires_at TEXT · NN · IDX
responded_at TEXT · NULL
created_at TEXT · NN · DEF
CLINIC_CLOSURES（诊所停诊安排）
PK id TEXT · NN
FK clinic_id → clinics TEXT · NN · IDX
starts_at TEXT · NN · IDX
ends_at TEXT · NN · IDX
reason_code TEXT · NN
reason_en TEXT · NN
reason_zh TEXT · NN
status TEXT · NN · DEF
created_at TEXT · NN · DEF
DENTIST_SCHEDULE_EXCEPTIONS（牙医排班例外）
PK id TEXT · NN
FK dentist_id → dentists TEXT · NN · IDX
FK clinic_id → clinics TEXT · NULL
starts_at TEXT · NN · IDX
ends_at TEXT · NN · IDX
exception_type TEXT · NN
reason_en TEXT · NN · DEF
reason_zh TEXT · NN · DEF
status TEXT · NN · DEF
created_at TEXT · NN · DEF
WAITLIST_ENTRIES（候补登记）
PK id TEXT · NN
FK patient_id → patients TEXT · NN
FK treatment_id → treatment_catalog TEXT · NN · IDX
FK preferred_clinic_id → clinics TEXT · NULL · IDX
earliest_date TEXT · NN · IDX
latest_date TEXT · NN · IDX
preferred_periods_json TEXT · NN · DEF
acceptable_clinic_ids_json TEXT · NN · DEF
notice_hours INTEGER · NN · DEF
priority_score INTEGER · NN · DEF
status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
临床就诊与病历版本
CLINICAL_VISITS（临床就诊）
PK id TEXT · NN
visit_code TEXT · NN · IDX
FK appointment_id → appointments TEXT · NN · IDX
FK patient_id → patients TEXT · NN · IDX
FK clinic_id → clinics TEXT · NN · IDX
FK dentist_id → dentists TEXT · NN
FK chair_id → clinic_chairs TEXT · NULL
status TEXT · NN · IDX · DEF
chief_complaint_en TEXT · NN · DEF
chief_complaint_zh TEXT · NN · DEF
safety_gate_status TEXT · NN · DEF
started_at TEXT · NN · IDX
completed_at TEXT · NULL
visit_version INTEGER · NN · DEF
FK created_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
CLINICAL_NOTES（临床记录）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
note_type TEXT · NN · IDX · DEF
current_version INTEGER · NN · DEF
status TEXT · NN · DEF
FK signed_by_dentist_id → dentists TEXT · NULL
signed_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
CLINICAL_ATTACHMENTS（临床附件）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
category TEXT · NN · IDX
file_name TEXT · NN
mime_type TEXT · NN
storage_key TEXT · NN · IDX
size_bytes INTEGER · NN
description_en TEXT · NN · DEF
description_zh TEXT · NN · DEF
content_checksum TEXT · NN
captured_at TEXT · NN
status TEXT · NN · DEF
created_at TEXT · NN · DEF
VISIT_ALLERGY_ACKNOWLEDGEMENTS（就诊过敏复核）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
FK allergy_id → patient_allergies TEXT · NN · IDX
FK acknowledged_by_user_id → app_users TEXT · NULL
acknowledged_at TEXT · NN · IDX
precaution_en TEXT · NN
precaution_zh TEXT · NN
outcome TEXT · NN · DEF
created_at TEXT · NN · DEF
TREATMENT_PLANS（治疗计划）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
plan_code TEXT · NN · IDX
status TEXT · NN · IDX · DEF
clinical_priority TEXT · NN · DEF
estimated_total_fen INTEGER · NN · DEF
acceptance_status TEXT · NN · DEF
accepted_at TEXT · NULL
accepted_by TEXT · NULL
valid_until TEXT · NULL
notes_en TEXT · NN · DEF
notes_zh TEXT · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
CLINICAL_NOTE_VERSIONS（临床记录版本）
PK id TEXT · NN
FK clinical_note_id → clinical_notes TEXT · NN · IDX
version_number INTEGER · NN · IDX
content_en TEXT · NN
content_zh TEXT · NN
amendment_reason TEXT · NULL
FK author_user_id → app_users TEXT · NULL
content_checksum TEXT · NN
created_at TEXT · NN · DEF
ODONTOGRAM_OBSERVATIONS（牙位观察）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
FK patient_id → patients TEXT · NN · IDX
tooth_code TEXT · NN · IDX
dentition TEXT · NN · DEF
surfaces_json TEXT · NN · DEF
observation_type TEXT · NN
severity TEXT · NN · DEF
finding_en TEXT · NN
finding_zh TEXT · NN
status TEXT · NN · DEF
FK observed_by_dentist_id → dentists TEXT · NN
observed_at TEXT · NN
supersedes_observation_id TEXT · NULL
created_at TEXT · NN · DEF
ACTUAL_TREATMENT_ITEMS（实际治疗项目）
PK id TEXT · NN
FK visit_id → clinical_visits TEXT · NN · IDX
FK requested_treatment_item_id → requested_t··· TEXT · NULL · IDX
FK treatment_id → treatment_catalog TEXT · NN
FK provider_dentist_id → dentists TEXT · NN
tooth_code TEXT · NULL
surfaces_json TEXT · NN · DEF
units INTEGER · NN · DEF
effective_unit_price_fen INTEGER · NN
line_amount_fen INTEGER · NN
price_source TEXT · NN · DEF
price_override_reason TEXT · NULL
notes_en TEXT · NN · DEF
notes_zh TEXT · NN · DEF
performed_at TEXT · NN · IDX
status TEXT · NN · DEF
created_at TEXT · NN · DEF
结算、收款与成本
FINANCIAL_REPORTING_PERIODS（财务报告期间）
PK id TEXT · NN
period_code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
starts_on TEXT · NN · IDX
ends_on TEXT · NN · IDX
status TEXT · NN · DEF
close_checklist_json TEXT · NN · DEF
locked_at TEXT · NULL
FK locked_by_user_id → app_users TEXT · NULL
reopened_at TEXT · NULL
FK reopened_by_user_id → app_users TEXT · NULL
reopen_reason TEXT · NULL
period_version INTEGER · NN · DEF
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PAYMENTS（收款流水）
PK id TEXT · NN
payment_code TEXT · NN · IDX
FK invoice_id → invoices TEXT · NN · IDX
amount_fen INTEGER · NN
method TEXT · NN
status TEXT · NN · DEF
simulated_gateway_reference TEXT · NN · IDX
received_at TEXT · NN · IDX
FK recorded_by_user_id → app_users TEXT · NULL
notes TEXT · NN · DEF
created_at TEXT · NN · DEF
INVOICES（诊疗结算单）
PK id TEXT · NN
invoice_code TEXT · NN · IDX
FK visit_id → clinical_visits TEXT · NN · IDX
FK patient_id → patients TEXT · NN
FK clinic_id → clinics TEXT · NN · IDX
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
status TEXT · NN · IDX · DEF
currency TEXT · NN · DEF
subtotal_fen INTEGER · NN · DEF
adjustment_total_fen INTEGER · NN · DEF
tax_total_fen INTEGER · NN · DEF
total_fen INTEGER · NN · DEF
paid_fen INTEGER · NN · DEF
refunded_fen INTEGER · NN · DEF
balance_fen INTEGER · NN · IDX · DEF
issued_at TEXT · NULL
due_at TEXT · NULL · IDX
voided_at TEXT · NULL
void_reason TEXT · NULL
invoice_version INTEGER · NN · DEF
FK created_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · DEF
PAYMENT_REFUNDS（退款流水）
PK id TEXT · NN
refund_code TEXT · NN · IDX
FK payment_id → payments TEXT · NN · IDX
amount_fen INTEGER · NN
reason_code TEXT · NN
reason_text TEXT · NN
status TEXT · NN · IDX · DEF
FK authorized_by_user_id → app_users TEXT · NULL
refunded_at TEXT · NN
created_at TEXT · NN · DEF
INVOICE_LINES（结算明细）
PK id TEXT · NN
FK invoice_id → invoices TEXT · NN · IDX
FK actual_treatment_item_id → actual_treatme··· TEXT · NN · IDX
line_number INTEGER · NN · IDX
treatment_code_snapshot TEXT · NN
description_en_snapshot TEXT · NN
description_zh_snapshot TEXT · NN
tooth_code_snapshot TEXT · NULL
surfaces_json_snapshot TEXT · NN · DEF
units_snapshot INTEGER · NN
unit_price_fen_snapshot INTEGER · NN
amount_fen_snapshot INTEGER · NN
cost_fen_snapshot INTEGER · NN · DEF
performed_at_snapshot TEXT · NN
created_at TEXT · NN · DEF
INVOICE_ADJUSTMENTS（结算调整）
PK id TEXT · NN
FK invoice_id → invoices TEXT · NN · IDX
adjustment_type TEXT · NN
amount_fen INTEGER · NN
reason_code TEXT · NN
reason_text TEXT · NN
status TEXT · NN · IDX · DEF
FK authored_by_user_id → app_users TEXT · NULL
FK approved_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
TREATMENT_COST_ENTRIES（治疗成本记录）
PK id TEXT · NN
FK actual_treatment_item_id → actual_treatme··· TEXT · NN · IDX
FK visit_id → clinical_visits TEXT · NN
FK clinic_id → clinics TEXT · NN · IDX
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
catalog_estimated_cost_fen INTEGER · NN
materials_cost_fen INTEGER · NN · DEF
laboratory_cost_fen INTEGER · NN · DEF
provider_cost_fen INTEGER · NN · DEF
total_cost_fen INTEGER · NN
cost_date TEXT · NN
source TEXT · NN · DEF
created_at TEXT · NN · DEF
期间核算与激励
OPERATING_EXPENSES（运营费用）
PK id TEXT · NN
FK clinic_id → clinics TEXT · NN · IDX
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
category TEXT · NN · IDX
vendor_name TEXT · NN · DEF
amount_fen INTEGER · NN
incurred_on TEXT · NN · IDX
description_en TEXT · NN
description_zh TEXT · NN
status TEXT · NN · DEF
FK entered_by_user_id → app_users TEXT · NULL
FK approved_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
INCENTIVE_POOLS（激励池）
PK id TEXT · NN
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
FK network_result_id → financial_period_resu··· TEXT · NN
rate_basis_points INTEGER · NN · DEF
eligible_net_profit_fen INTEGER · NN
pool_amount_fen INTEGER · NN
formula_evidence TEXT · NN
status TEXT · NN · DEF
approved_at TEXT · NULL
FK approved_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
FINANCIAL_CLOSE_EXCEPTIONS（期间关闭异常）
PK id TEXT · NN
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
FK clinic_id → clinics TEXT · NULL
exception_type TEXT · NN
severity TEXT · NN · IDX
entity_type TEXT · NN
entity_id TEXT · NN
description_en TEXT · NN
description_zh TEXT · NN
status TEXT · NN · IDX · DEF
resolved_at TEXT · NULL
resolution_note TEXT · NULL
created_at TEXT · NN · DEF
FINANCIAL_PERIOD_RESULTS（期间核算结果）
PK id TEXT · NN
FK reporting_period_id → financial_reporting··· TEXT · NN · IDX
FK clinic_id → clinics TEXT · NULL
scope_key TEXT · NN · IDX
recognized_revenue_fen INTEGER · NN
treatment_cost_fen INTEGER · NN
operating_expense_fen INTEGER · NN
net_profit_fen INTEGER · NN · IDX
net_margin_basis_points INTEGER · NN
invoice_count INTEGER · NN · DEF
open_exception_count INTEGER · NN · DEF
calculation_inputs_json TEXT · NN
status TEXT · NN · DEF
calculated_at TEXT · NN
approved_at TEXT · NULL
FK approved_by_user_id → app_users TEXT · NULL
created_at TEXT · NN · DEF
扩张情景与候选诊所
EXPANSION_SCENARIOS（扩张情景）
PK id TEXT · NN
scenario_code TEXT · NN · IDX
version INTEGER · NN · IDX · DEF
name_en TEXT · NN
name_zh TEXT · NN
description_en TEXT · NN
description_zh TEXT · NN
forecast_horizon_months INTEGER · NN · DEF
status TEXT · NN · IDX · DEF
advisory_disclaimer_en TEXT · NN
advisory_disclaimer_zh TEXT · NN
FK created_by_user_id → app_users TEXT · NULL
FK approved_by_user_id → app_users TEXT · NULL
approved_at TEXT · NULL
created_at TEXT · NN · DEF
updated_at TEXT · NN · IDX · DEF
EXPANSION_CANDIDATE_CLINICS（候选诊所）
PK id TEXT · NN
FK scenario_id → expansion_scenarios TEXT · NN · IDX
candidate_code TEXT · NN · IDX
name_en TEXT · NN
name_zh TEXT · NN
city_en TEXT · NN
city_zh TEXT · NN
district_en TEXT · NN
district_zh TEXT · NN
planned_chair_count INTEGER · NN
planned_dentist_count INTEGER · NN
planned_support_staff_count INTEGER · NN
forecast_quarterly_appointments INTEGER · NN
initial_capital_fen INTEGER · NN
quarterly_fixed_cost_fen INTEGER · NN
required_qualifications_json TEXT · NN · DEF
qualification_coverage_basis_points INTEGER · NN
travel_constraint_minutes INTEGER · NN
readiness_status TEXT · NN · IDX · DEF
created_at TEXT · NN · DEF
EXPANSION_ASSUMPTIONS（扩张假设）
PK id TEXT · NN
FK scenario_id → expansion_scenarios TEXT · NN · IDX
assumption_key TEXT · NN · IDX
label_en TEXT · NN
label_zh TEXT · NN
value_text TEXT · NN
unit TEXT · NN
source_en TEXT · NN
source_zh TEXT · NN
created_at TEXT · NN · DEF
EXPANSION_SCENARIO_RESULTS（扩张情景结果）
PK id TEXT · NN
FK scenario_id → expansion_scenarios TEXT · NN · IDX
projected_quarter_revenue_fen INTEGER · NN
projected_treatment_cost_fen INTEGER · NN
projected_operating_expense_fen INTEGER · NN
projected_net_profit_fen INTEGER · NN
projected_margin_basis_points INTEGER · NN
payback_months INTEGER · NN
demand_coverage_basis_points INTEGER · NN
qualification_coverage_basis_points INTEGER · NN
composite_score INTEGER · NN · IDX
risk_level TEXT · NN · IDX
formula_evidence_json TEXT · NN
calculated_at TEXT · NN
created_at TEXT · NN · DEF

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 组织、账户与授权



### ORGANIZATIONS（组织）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | timezone | TEXT · NN · DEF |
| | currency | TEXT · NN · DEF |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### ACCOUNT_INVITATIONS（账户邀请）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | phone_e164 | TEXT · NN · IDX |
| FK | target_user_id → app_users | TEXT · NULL |
| | role_id | TEXT · NULL |
| | clinic_id | TEXT · NULL |
| | status | TEXT · NN · IDX · DEF |
| | expires_at | TEXT · NN |
| | accepted_at | TEXT · NULL |
| FK | created_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### ROLE_PERMISSIONS（角色权限）

| | | |
|---|---|---|
| PK/FK | role_id → roles | TEXT · NN |
| PK/FK | permission_id → permissions | TEXT · NN · IDX |
| | created_at | TEXT · NN · DEF |

### AUDIT_EVENTS（审计事件）

| | | |
|---|---|---|
| PK | id | INTEGER · NN |
| | request_id | TEXT · NN · IDX |
| FK | actor_user_id → app_users | TEXT · NULL · IDX |
| | actor_subject | TEXT · NULL |
| | action | TEXT · NN · IDX |
| | entity_type | TEXT · NN · IDX |
| | entity_id | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NULL · IDX |
| | outcome | TEXT · NN · DEF |
| | source | TEXT · NN · DEF |
| | reason | TEXT · NULL |
| | before_json | TEXT · NULL |
| | after_json | TEXT · NULL |
| | metadata_json | TEXT · NULL |
| | occurred_at | TEXT · NN · IDX · DEF |

### CLINICS（诊所）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | organization_id → organizations | TEXT · NN · IDX |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | city_en | TEXT · NN |
| | city_zh | TEXT · NN |
| | address_en | TEXT · NN · DEF |
| | address_zh | TEXT · NN · DEF |
| | phone | TEXT · NULL |
| | email | TEXT · NULL |
| | chair_capacity | INTEGER · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | opened_on | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### ROLES（角色）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | description_en | TEXT · NN · DEF |
| | description_zh | TEXT · NN · DEF |
| | is_system | INTEGER · NN · DEF |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### SYSTEM_SETTINGS（系统配置）

| | | |
|---|---|---|
| PK/FK | organization_id → organizations | TEXT · NN |
| PK | setting_key | TEXT · NN |
| | value_json | TEXT · NN |
| FK | updated_by_user_id → app_users | TEXT · NULL |
| | updated_at | TEXT · NN · DEF |

### APP_USERS（系统账户）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | auth_subject | TEXT · NN · IDX |
| | email | TEXT · NN · IDX |
| | display_name | TEXT · NN |
| | phone_e164 | TEXT · NULL · IDX |
| | username | TEXT · NULL · IDX |
| | password_hash | TEXT · NULL |
| | password_salt | TEXT · NULL |
| | password_iterations | INTEGER · NULL |
| | password_set_at | TEXT · NULL |
| | last_auth_method | TEXT · NULL |
| | failed_login_count | INTEGER · NN · DEF |
| | locked_until | TEXT · NULL |
| | preferred_language | TEXT · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | last_sign_in_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### PERMISSIONS（权限）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | code | TEXT · NN · IDX |
| | domain | TEXT · NN · IDX |
| | description_en | TEXT · NN |
| | description_zh | TEXT · NN |
| | created_at | TEXT · NN · DEF |

### USER_ROLE_ASSIGNMENTS（用户角色授权）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | user_id → app_users | TEXT · NN · IDX |
| FK | role_id → roles | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NULL · IDX |
| | status | TEXT · NN · IDX · DEF |
| | effective_from | TEXT · NN |
| | effective_to | TEXT · NULL |
| FK | granted_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 身份认证与通知

### AUTH_IDENTITIES（外部身份）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | user_id → app_users | TEXT · NN · IDX |
| | provider | TEXT · NN · IDX |
| | provider_subject | TEXT · NN · IDX |
| | profile_json | TEXT · NN · DEF |
| | last_used_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### AUTH_OAUTH_STATES（OAuth 状态）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | provider | TEXT · NN |
| | state_hash | TEXT · NN · IDX |
| | return_to | TEXT · NN · DEF |
| | expires_at | TEXT · NN · IDX |
| | consumed_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### AUTH_OTP_CHALLENGES（验证码挑战）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | phone_e164 | TEXT · NN · IDX |
| | purpose | TEXT · NN · IDX |
| | code_hash | TEXT · NN |
| | code_salt | TEXT · NN |
| | expires_at | TEXT · NN · IDX |
| | consumed_at | TEXT · NULL |
| | attempt_count | INTEGER · NN · DEF |
| | max_attempts | INTEGER · NN · DEF |
| | request_ip_hash | TEXT · NULL |
| | created_at | TEXT · NN · IDX · DEF |

### NOTIFICATION_EVENTS（通知事件）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | appointment_id → appointments | TEXT · NULL · IDX |
| | channel | TEXT · NN |
| | template_code | TEXT · NN |
| | recipient_masked | TEXT · NN |
| | locale | TEXT · NN · DEF |
| | status | TEXT · NN · DEF |
| | idempotency_key | TEXT · NN · IDX |
| | queued_at | TEXT · NN · IDX |
| | sent_at | TEXT · NULL |
| | delivered_at | TEXT · NULL |
| | failure_code | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

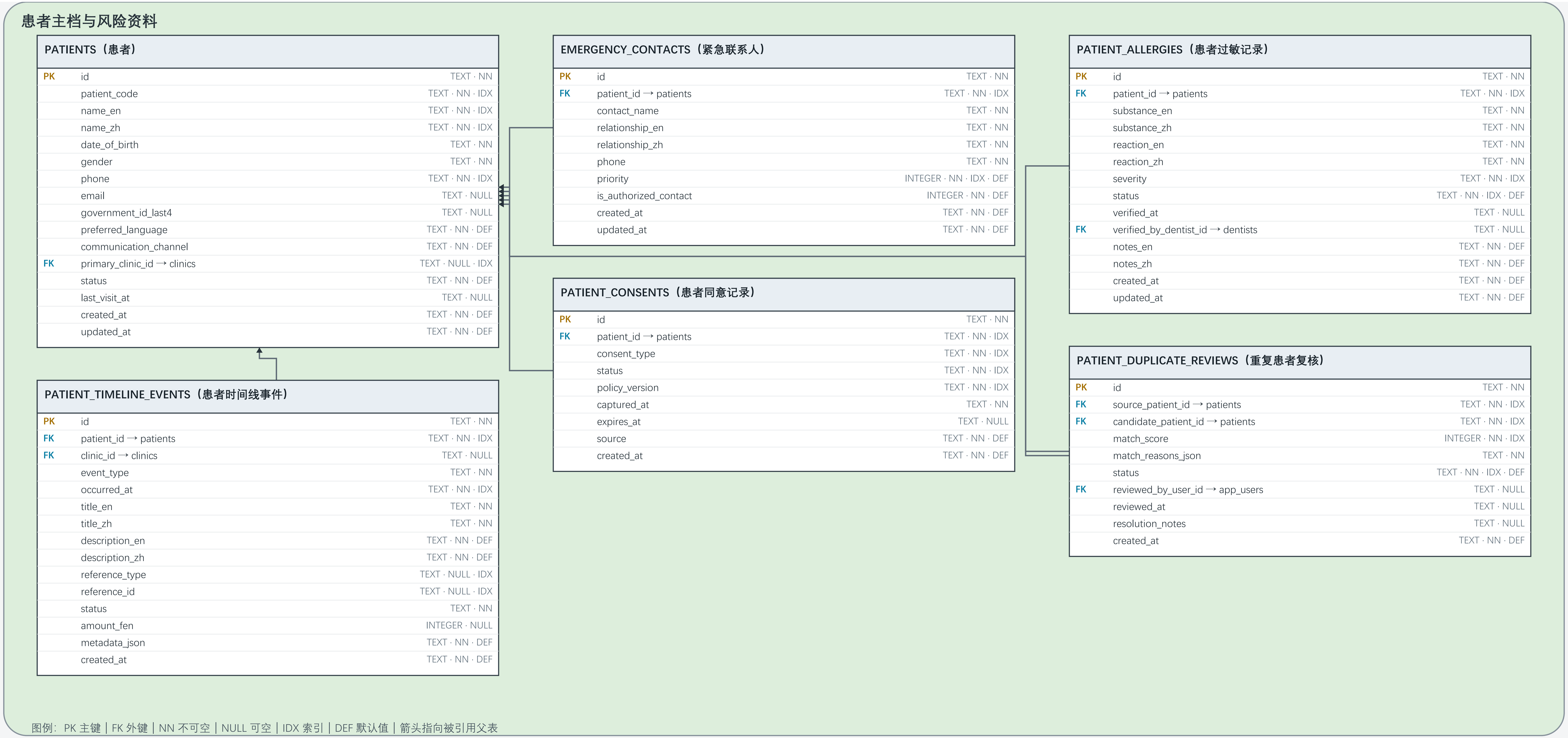
### AUTH_SESSIONS（认证会话）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | user_id → app_users | TEXT · NN · IDX |
| | token_hash | TEXT · NN · IDX |
| | auth_method | TEXT · NN |
| | expires_at | TEXT · NN · IDX |
| | revoked_at | TEXT · NULL |
| | last_seen_at | TEXT · NULL |
| | ip_hash | TEXT · NULL |
| | user_agent_hash | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### PATIENT_ACCOUNT_LINKS（患者账户绑定）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | user_id → app_users | TEXT · NN · IDX |
| FK | patient_id → patients | TEXT · NN · IDX |
| | verification_method | TEXT · NN |
| | verified_at | TEXT · NN |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表



## 患者主档与风险资料

### PATIENTS（患者）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | patient_code | TEXT · NN · IDX |
| | name_en | TEXT · NN · IDX |
| | name_zh | TEXT · NN · IDX |
| | date_of_birth | TEXT · NN |
| | gender | TEXT · NN |
| | phone | TEXT · NN · IDX |
| | email | TEXT · NULL |
| | government_id_last4 | TEXT · NULL |
| | preferred_language | TEXT · NN · DEF |
| | communication_channel | TEXT · NN · DEF |
| FK | primary_clinic_id → clinics | TEXT · NULL · IDX |
| | status | TEXT · NN · DEF |
| | last_visit_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### EMERGENCY_CONTACTS（紧急联系人）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | patient_id → patients | TEXT · NN · IDX |
| | contact_name | TEXT · NN |
| | relationship_en | TEXT · NN |
| | relationship_zh | TEXT · NN |
| | phone | TEXT · NN |
| | priority | INTEGER · NN · IDX · DEF |
| | is_authorized_contact | INTEGER · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### PATIENT_ALLERGIES（患者过敏记录）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | patient_id → patients | TEXT · NN · IDX |
| | substance_en | TEXT · NN |
| | substance_zh | TEXT · NN |
| | reaction_en | TEXT · NN |
| | reaction_zh | TEXT · NN |
| | severity | TEXT · NN · IDX |
| | status | TEXT · NN · IDX · DEF |
| | verified_at | TEXT · NULL |
| FK | verified_by_dentist_id → dentists | TEXT · NULL |
| | notes_en | TEXT · NN · DEF |
| | notes_zh | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### PATIENT_CONSENTS（患者同意记录）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | patient_id → patients | TEXT · NN · IDX |
| | consent_type | TEXT · NN · IDX |
| | status | TEXT · NN · IDX |
| | policy_version | TEXT · NN · IDX |
| | captured_at | TEXT · NN |
| | expires_at | TEXT · NULL |
| | source | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### PATIENT_TIMELINE_EVENTS（患者时间线事件）

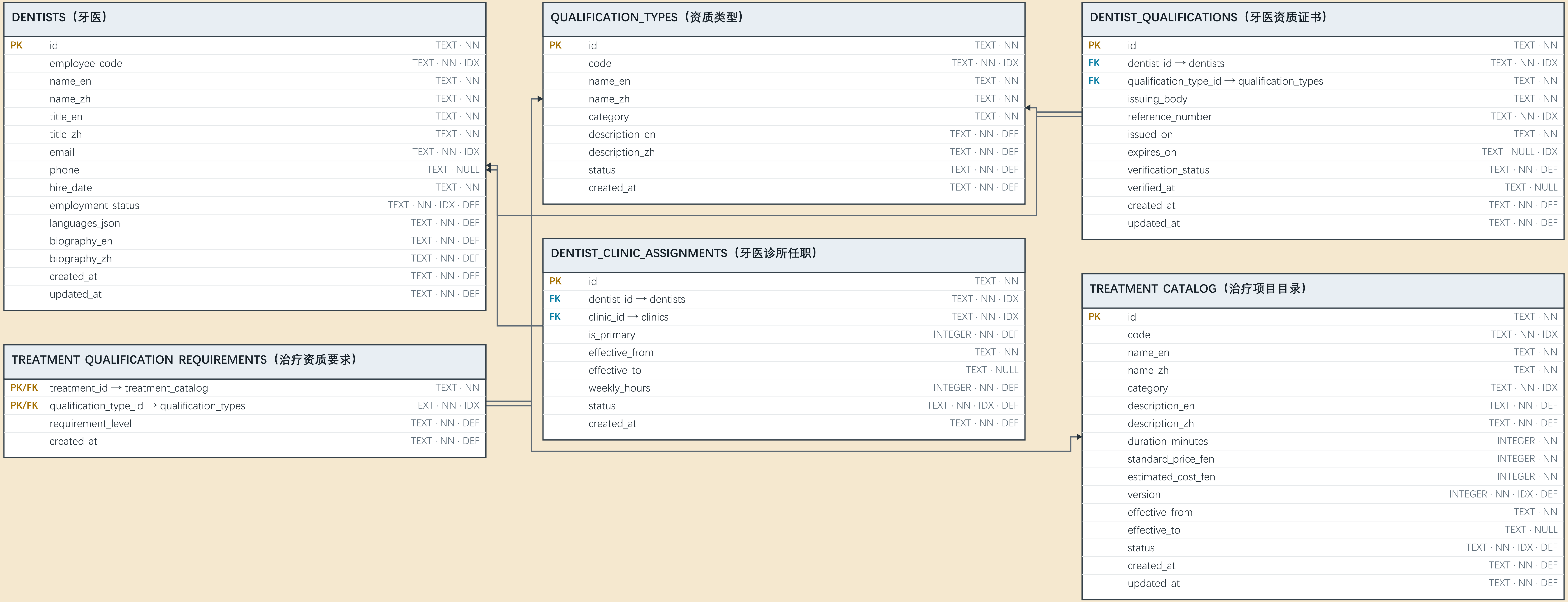
| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | patient_id → patients | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NULL |
| | event_type | TEXT · NN |
| | occurred_at | TEXT · NN · IDX |
| | title_en | TEXT · NN |
| | title_zh | TEXT · NN |
| | description_en | TEXT · NN · DEF |
| | description_zh | TEXT · NN · DEF |
| | reference_type | TEXT · NULL · IDX |
| | reference_id | TEXT · NULL · IDX |
| | status | TEXT · NN |
| | amount_fen | INTEGER · NULL |
| | metadata_json | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### PATIENT_DUPLICATE_REVIEWS（重复患者复核）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | source_patient_id → patients | TEXT · NN · IDX |
| FK | candidate_patient_id → patients | TEXT · NN · IDX |
| | match_score | INTEGER · NN · IDX |
| | match_reasons_json | TEXT · NN |
| | status | TEXT · NN · IDX · DEF |
| FK | reviewed_by_user_id → app_users | TEXT · NULL |
| | reviewed_at | TEXT · NULL |
| | resolution_notes | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 牙医、资质与治疗目录



### DENTISTS（牙医）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | employee_code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | title_en | TEXT · NN |
| | title_zh | TEXT · NN |
| | email | TEXT · NN · IDX |
| | phone | TEXT · NULL |
| | hire_date | TEXT · NN |
| | employment_status | TEXT · NN · IDX · DEF |
| | languages_json | TEXT · NN · DEF |
| | biography_en | TEXT · NN · DEF |
| | biography_zh | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### TREATMENT_QUALIFICATION_REQUIREMENTS（治疗资质要求）

| | | |
|---|---|---|
| PK/FK | treatment_id → treatment_catalog | TEXT · NN |
| PK/FK | qualification_type_id → qualification_types | TEXT · NN · IDX |
| | requirement_level | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### QUALIFICATION_TYPES（资质类型）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | category | TEXT · NN |
| | description_en | TEXT · NN · DEF |
| | description_zh | TEXT · NN · DEF |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### DENTIST_CLINIC_ASSIGNMENTS（牙医诊所任职）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | dentist_id → dentists | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| | is_primary | INTEGER · NN · DEF |
| | effective_from | TEXT · NN |
| | effective_to | TEXT · NULL |
| | weekly_hours | INTEGER · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |

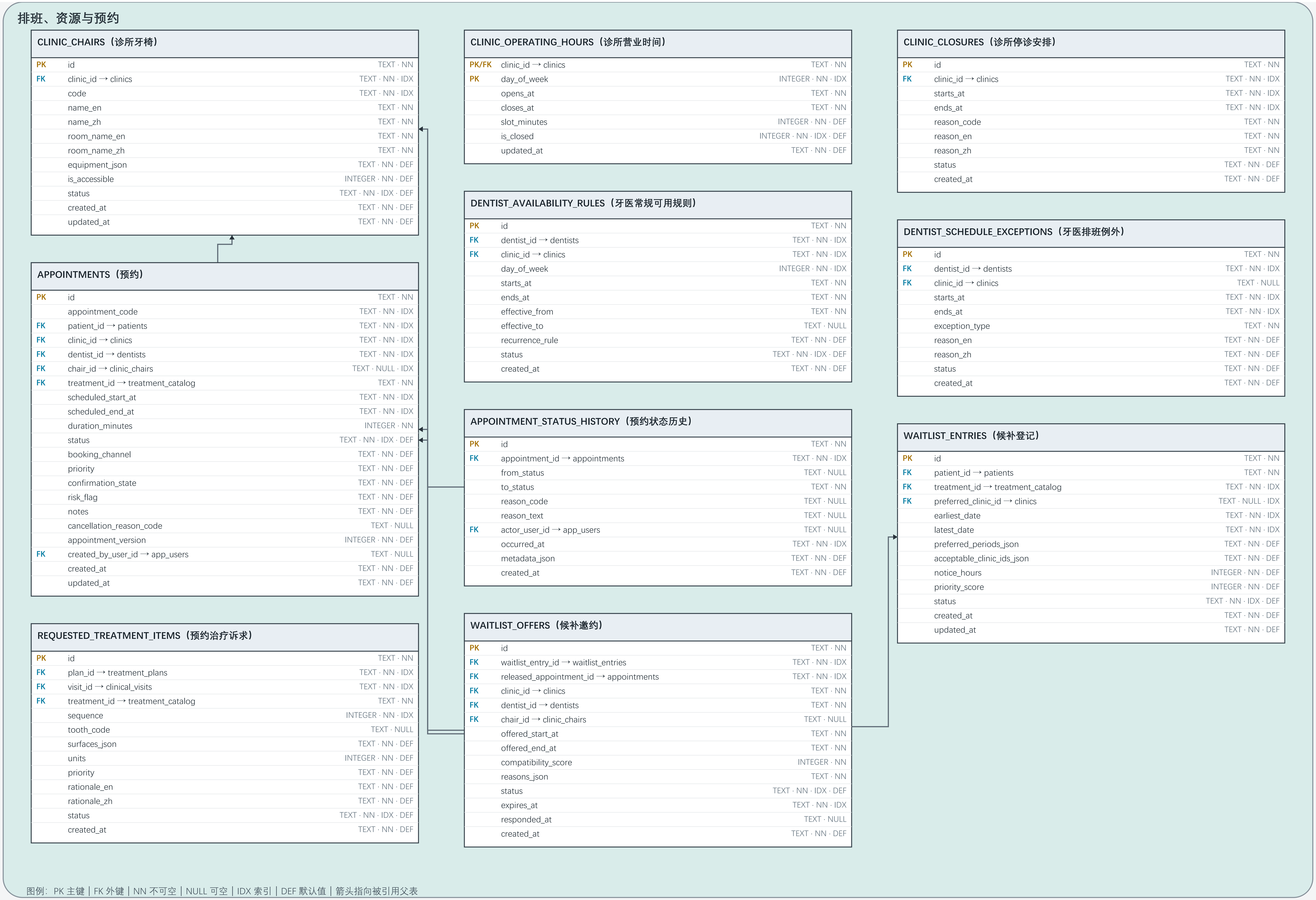
### DENTIST_QUALIFICATIONS（牙医资质证书）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | dentist_id → dentists | TEXT · NN · IDX |
| FK | qualification_type_id → qualification_types | TEXT · NN |
| | issuing_body | TEXT · NN |
| | reference_number | TEXT · NN · IDX |
| | issued_on | TEXT · NN |
| | expires_on | TEXT · NULL · IDX |
| | verification_status | TEXT · NN · DEF |
| | verified_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### TREATMENT_CATALOG（治疗项目目录）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | category | TEXT · NN · IDX |
| | description_en | TEXT · NN · DEF |
| | description_zh | TEXT · NN · DEF |
| | duration_minutes | INTEGER · NN |
| | standard_price_fen | INTEGER · NN |
| | estimated_cost_fen | INTEGER · NN |
| | version | INTEGER · NN · IDX · DEF |
| | effective_from | TEXT · NN |
| | effective_to | TEXT · NULL |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表



## 排班、资源与预约

### CLINIC_CHAIRS（诊所牙椅）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| | code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | room_name_en | TEXT · NN |
| | room_name_zh | TEXT · NN |
| | equipment_json | TEXT · NN · DEF |
| | is_accessible | INTEGER · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### APPOINTMENTS（预约）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | appointment_code | TEXT · NN · IDX |
| FK | patient_id → patients | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| FK | dentist_id → dentists | TEXT · NN · IDX |
| FK | chair_id → clinic_chairs | TEXT · NULL · IDX |
| FK | treatment_id → treatment_catalog | TEXT · NN |
| | scheduled_start_at | TEXT · NN · IDX |
| | scheduled_end_at | TEXT · NN · IDX |
| | duration_minutes | INTEGER · NN |
| | status | TEXT · NN · IDX · DEF |
| | booking_channel | TEXT · NN · DEF |
| | priority | TEXT · NN · DEF |
| | confirmation_state | TEXT · NN · DEF |
| | risk_flag | TEXT · NN · DEF |
| | notes | TEXT · NN · DEF |
| | cancellation_reason_code | TEXT · NULL |
| | appointment_version | INTEGER · NN · DEF |
| FK | created_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### REQUESTED_TREATMENT_ITEMS（预约治疗诉求）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | plan_id → treatment_plans | TEXT · NN · IDX |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| FK | treatment_id → treatment_catalog | TEXT · NN |
| | sequence | INTEGER · NN · IDX |
| | tooth_code | TEXT · NULL |
| | surfaces_json | TEXT · NN · DEF |
| | units | INTEGER · NN · DEF |
| | priority | TEXT · NN · DEF |
| | rationale_en | TEXT · NN · DEF |
| | rationale_zh | TEXT · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |

### CLINIC_OPERATING_HOURS（诊所营业时间）

| | | |
|---|---|---|
| PK/FK | clinic_id → clinics | TEXT · NN |
| PK | day_of_week | INTEGER · NN · IDX |
| | opens_at | TEXT · NN |
| | closes_at | TEXT · NN |
| | slot_minutes | INTEGER · NN · DEF |
| | is_closed | INTEGER · NN · IDX · DEF |
| | updated_at | TEXT · NN · DEF |

### DENTIST_AVAILABILITY_RULES（牙医常规可用规则）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | dentist_id → dentists | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| | day_of_week | INTEGER · NN · IDX |
| | starts_at | TEXT · NN |
| | ends_at | TEXT · NN |
| | effective_from | TEXT · NN |
| | effective_to | TEXT · NULL |
| | recurrence_rule | TEXT · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |

### APPOINTMENT_STATUS_HISTORY（预约状态历史）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | appointment_id → appointments | TEXT · NN · IDX |
| | from_status | TEXT · NULL |
| | to_status | TEXT · NN |
| | reason_code | TEXT · NULL |
| | reason_text | TEXT · NULL |
| FK | actor_user_id → app_users | TEXT · NULL |
| | occurred_at | TEXT · NN · IDX |
| | metadata_json | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### WAITLIST_OFFERS（候补邀约）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | waitlist_entry_id → waitlist_entries | TEXT · NN · IDX |
| FK | released_appointment_id → appointments | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NN |
| FK | dentist_id → dentists | TEXT · NN |
| FK | chair_id → clinic_chairs | TEXT · NULL |
| | offered_start_at | TEXT · NN |
| | offered_end_at | TEXT · NN |
| | compatibility_score | INTEGER · NN |
| | reasons_json | TEXT · NN |
| | status | TEXT · NN · IDX · DEF |
| | expires_at | TEXT · NN · IDX |
| | responded_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### CLINIC_CLOSURES（诊所停诊安排）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| | starts_at | TEXT · NN · IDX |
| | ends_at | TEXT · NN · IDX |
| | reason_code | TEXT · NN |
| | reason_en | TEXT · NN |
| | reason_zh | TEXT · NN |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### DENTIST_SCHEDULE_EXCEPTIONS（牙医排班例外）

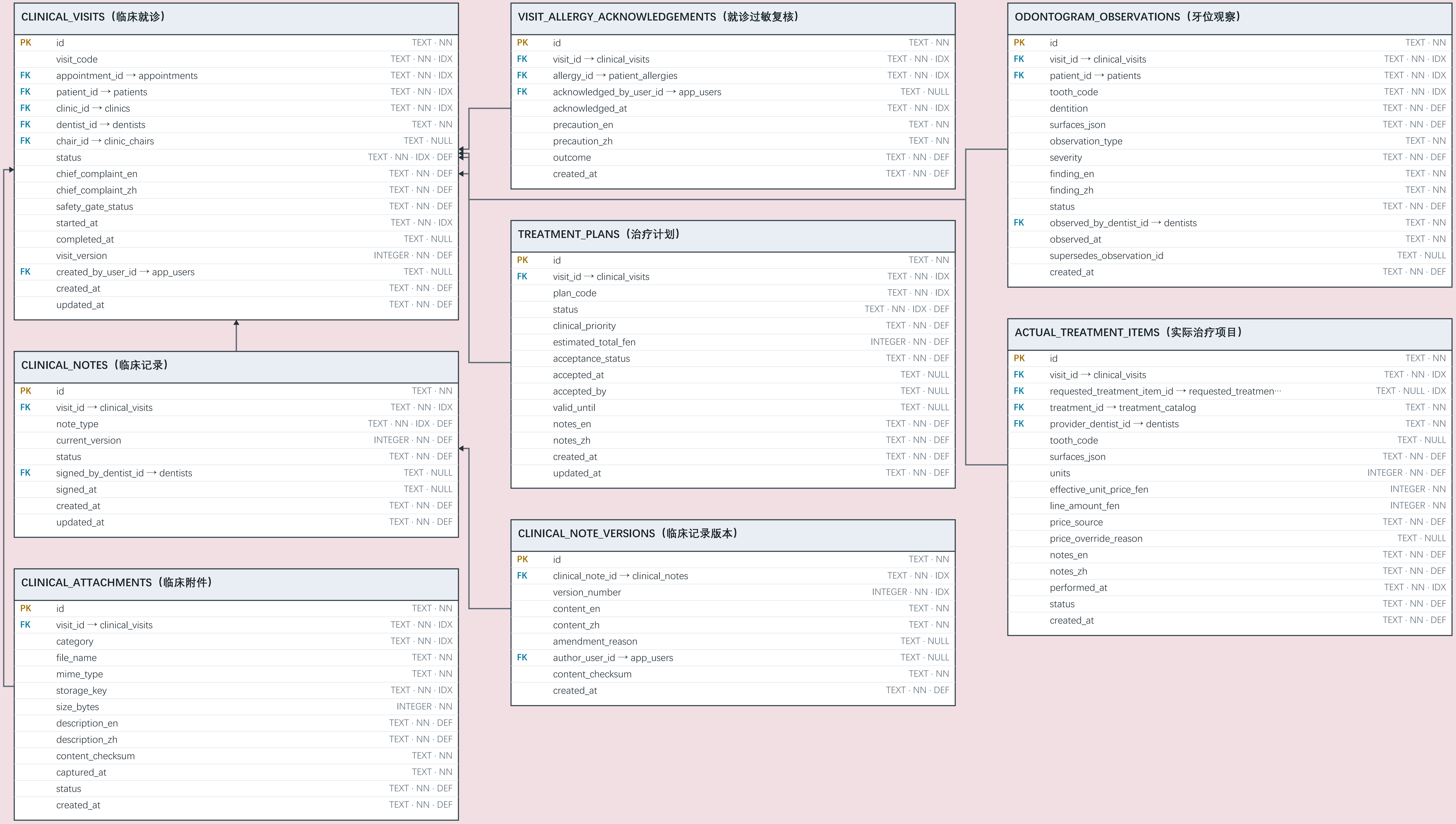
| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | dentist_id → dentists | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NULL |
| | starts_at | TEXT · NN · IDX |
| | ends_at | TEXT · NN · IDX |
| | exception_type | TEXT · NN |
| | reason_en | TEXT · NN · DEF |
| | reason_zh | TEXT · NN · DEF |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### WAITLIST_ENTRIES（候补登记）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | patient_id → patients | TEXT · NN |
| FK | treatment_id → treatment_catalog | TEXT · NN · IDX |
| FK | preferred_clinic_id → clinics | TEXT · NULL · IDX |
| | earliest_date | TEXT · NN · IDX |
| | latest_date | TEXT · NN · IDX |
| | preferred_periods_json | TEXT · NN · DEF |
| | acceptable_clinic_ids_json | TEXT · NN · DEF |
| | notice_hours | INTEGER · NN · DEF |
| | priority_score | INTEGER · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 临床就诊与病历版本



### CLINICAL_VISITS（临床就诊）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | visit_code | TEXT · NN · IDX |
| FK | appointment_id → appointments | TEXT · NN · IDX |
| FK | patient_id → patients | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| FK | dentist_id → dentists | TEXT · NN |
| FK | chair_id → clinic_chairs | TEXT · NULL |
| | status | TEXT · NN · IDX · DEF |
| | chief_complaint_en | TEXT · NN · DEF |
| | chief_complaint_zh | TEXT · NN · DEF |
| | safety_gate_status | TEXT · NN · DEF |
| | started_at | TEXT · NN · IDX |
| | completed_at | TEXT · NULL |
| | visit_version | INTEGER · NN · DEF |
| FK | created_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### VISIT_ALLERGY_ACKNOWLEDGEMENTS（就诊过敏复核）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| FK | allergy_id → patient_allergies | TEXT · NN · IDX |
| FK | acknowledged_by_user_id → app_users | TEXT · NULL |
| | acknowledged_at | TEXT · NN · IDX |
| | precaution_en | TEXT · NN |
| | precaution_zh | TEXT · NN |
| | outcome | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### ODONTOGRAM_OBSERVATIONS（牙位观察）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| FK | patient_id → patients | TEXT · NN · IDX |
| | tooth_code | TEXT · NN · IDX |
| | dentition | TEXT · NN · DEF |
| | surfaces_json | TEXT · NN · DEF |
| | observation_type | TEXT · NN |
| | severity | TEXT · NN · DEF |
| | finding_en | TEXT · NN |
| | finding_zh | TEXT · NN |
| | status | TEXT · NN · DEF |
| FK | observed_by_dentist_id → dentists | TEXT · NN |
| | observed_at | TEXT · NN |
| | supersedes_observation_id | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### TREATMENT_PLANS（治疗计划）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| | plan_code | TEXT · NN · IDX |
| | status | TEXT · NN · IDX · DEF |
| | clinical_priority | TEXT · NN · DEF |
| | estimated_total_fen | INTEGER · NN · DEF |
| | acceptance_status | TEXT · NN · DEF |
| | accepted_at | TEXT · NULL |
| | accepted_by | TEXT · NULL |
| | valid_until | TEXT · NULL |
| | notes_en | TEXT · NN · DEF |
| | notes_zh | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### CLINICAL_NOTES（临床记录）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| | note_type | TEXT · NN · IDX · DEF |
| | current_version | INTEGER · NN · DEF |
| | status | TEXT · NN · DEF |
| FK | signed_by_dentist_id → dentists | TEXT · NULL |
| | signed_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### ACTUAL_TREATMENT_ITEMS（实际治疗项目）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| FK | requested_treatment_item_id → requested_treatmen··· | TEXT · NULL · IDX |
| FK | treatment_id → treatment_catalog | TEXT · NN |
| FK | provider_dentist_id → dentists | TEXT · NN |
| | tooth_code | TEXT · NULL |
| | surfaces_json | TEXT · NN · DEF |
| | units | INTEGER · NN · DEF |
| | effective_unit_price_fen | INTEGER · NN |
| | line_amount_fen | INTEGER · NN |
| | price_source | TEXT · NN · DEF |
| | price_override_reason | TEXT · NULL |
| | notes_en | TEXT · NN · DEF |
| | notes_zh | TEXT · NN · DEF |
| | performed_at | TEXT · NN · IDX |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### CLINICAL_NOTE_VERSIONS（临床记录版本）

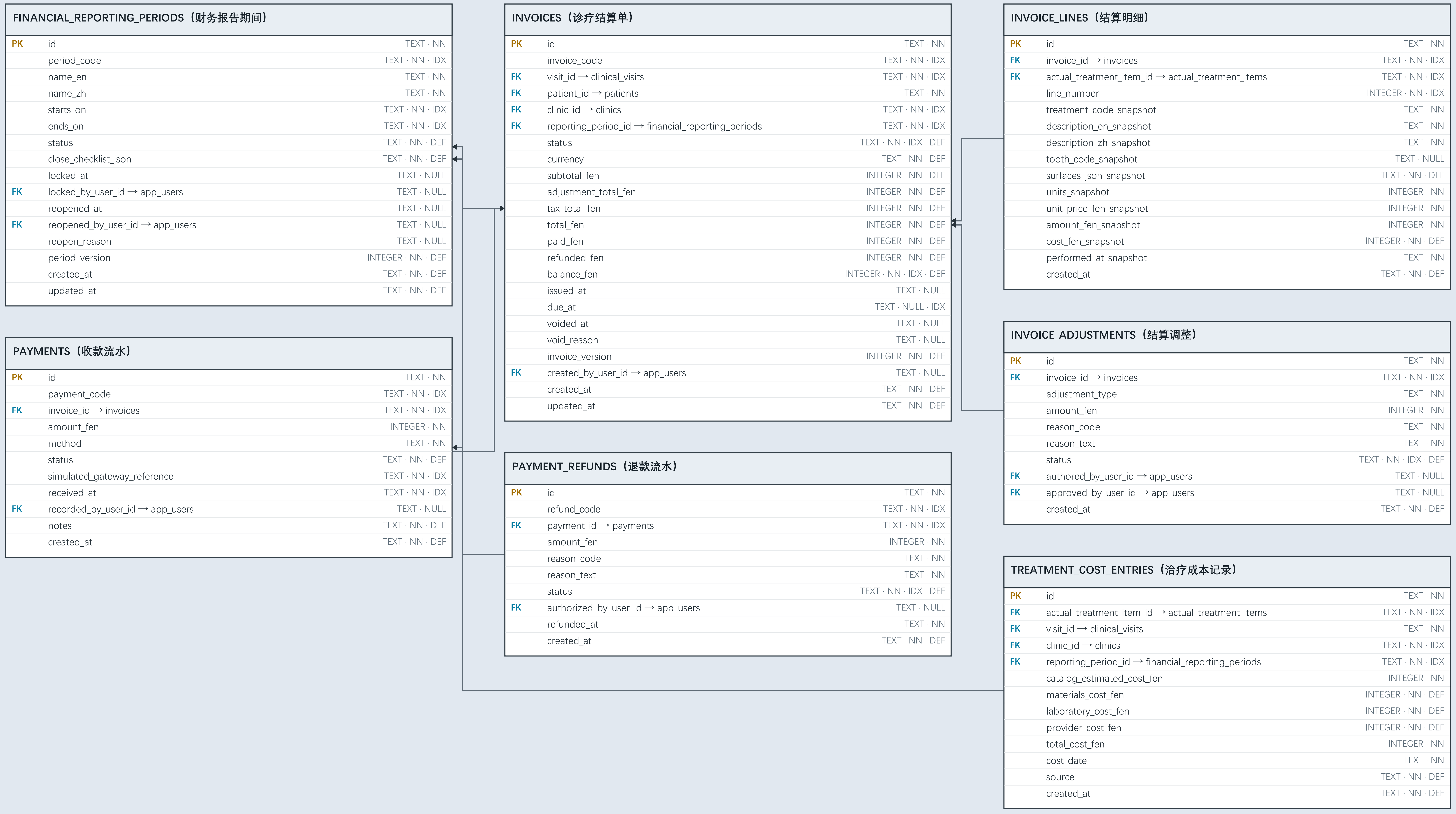
| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | clinical_note_id → clinical_notes | TEXT · NN · IDX |
| | version_number | INTEGER · NN · IDX |
| | content_en | TEXT · NN |
| | content_zh | TEXT · NN |
| | amendment_reason | TEXT · NULL |
| FK | author_user_id → app_users | TEXT · NULL |
| | content_checksum | TEXT · NN |
| | created_at | TEXT · NN · DEF |

### CLINICAL_ATTACHMENTS（临床附件）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| | category | TEXT · NN · IDX |
| | file_name | TEXT · NN |
| | mime_type | TEXT · NN |
| | storage_key | TEXT · NN · IDX |
| | size_bytes | INTEGER · NN |
| | description_en | TEXT · NN · DEF |
| | description_zh | TEXT · NN · DEF |
| | content_checksum | TEXT · NN |
| | captured_at | TEXT · NN |
| | status | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 结算、收款与成本



### FINANCIAL_REPORTING_PERIODS（财务报告期间）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | period_code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | starts_on | TEXT · NN · IDX |
| | ends_on | TEXT · NN · IDX |
| | status | TEXT · NN · DEF |
| | close_checklist_json | TEXT · NN · DEF |
| | locked_at | TEXT · NULL |
| FK | locked_by_user_id → app_users | TEXT · NULL |
| | reopened_at | TEXT · NULL |
| FK | reopened_by_user_id → app_users | TEXT · NULL |
| | reopen_reason | TEXT · NULL |
| | period_version | INTEGER · NN · DEF |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### INVOICES（诊疗结算单）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | invoice_code | TEXT · NN · IDX |
| FK | visit_id → clinical_visits | TEXT · NN · IDX |
| FK | patient_id → patients | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| | status | TEXT · NN · IDX · DEF |
| | currency | TEXT · NN · DEF |
| | subtotal_fen | INTEGER · NN · DEF |
| | adjustment_total_fen | INTEGER · NN · DEF |
| | tax_total_fen | INTEGER · NN · DEF |
| | total_fen | INTEGER · NN · DEF |
| | paid_fen | INTEGER · NN · DEF |
| | refunded_fen | INTEGER · NN · DEF |
| | balance_fen | INTEGER · NN · IDX · DEF |
| | issued_at | TEXT · NULL |
| | due_at | TEXT · NULL · IDX |
| | voided_at | TEXT · NULL |
| | void_reason | TEXT · NULL |
| | invoice_version | INTEGER · NN · DEF |
| FK | created_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · DEF |

### INVOICE_LINES（结算明细）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | invoice_id → invoices | TEXT · NN · IDX |
| FK | actual_treatment_item_id → actual_treatment_items | TEXT · NN · IDX |
| | line_number | INTEGER · NN · IDX |
| | treatment_code_snapshot | TEXT · NN |
| | description_en_snapshot | TEXT · NN |
| | description_zh_snapshot | TEXT · NN |
| | tooth_code_snapshot | TEXT · NULL |
| | surfaces_json_snapshot | TEXT · NN · DEF |
| | units_snapshot | INTEGER · NN |
| | unit_price_fen_snapshot | INTEGER · NN |
| | amount_fen_snapshot | INTEGER · NN |
| | cost_fen_snapshot | INTEGER · NN · DEF |
| | performed_at_snapshot | TEXT · NN |
| | created_at | TEXT · NN · DEF |

### PAYMENTS（收款流水）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | payment_code | TEXT · NN · IDX |
| FK | invoice_id → invoices | TEXT · NN · IDX |
| | amount_fen | INTEGER · NN |
| | method | TEXT · NN |
| | status | TEXT · NN · DEF |
| | simulated_gateway_reference | TEXT · NN · IDX |
| | received_at | TEXT · NN · IDX |
| FK | recorded_by_user_id → app_users | TEXT · NULL |
| | notes | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

### INVOICE_ADJUSTMENTS（结算调整）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | invoice_id → invoices | TEXT · NN · IDX |
| | adjustment_type | TEXT · NN |
| | amount_fen | INTEGER · NN |
| | reason_code | TEXT · NN |
| | reason_text | TEXT · NN |
| | status | TEXT · NN · IDX · DEF |
| FK | authored_by_user_id → app_users | TEXT · NULL |
| FK | approved_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### PAYMENT_REFUNDS（退款流水）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | refund_code | TEXT · NN · IDX |
| FK | payment_id → payments | TEXT · NN · IDX |
| | amount_fen | INTEGER · NN |
| | reason_code | TEXT · NN |
| | reason_text | TEXT · NN |
| | status | TEXT · NN · IDX · DEF |
| FK | authorized_by_user_id → app_users | TEXT · NULL |
| | refunded_at | TEXT · NN |
| | created_at | TEXT · NN · DEF |

### TREATMENT_COST_ENTRIES（治疗成本记录）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | actual_treatment_item_id → actual_treatment_items | TEXT · NN · IDX |
| FK | visit_id → clinical_visits | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| | catalog_estimated_cost_fen | INTEGER · NN |
| | materials_cost_fen | INTEGER · NN · DEF |
| | laboratory_cost_fen | INTEGER · NN · DEF |
| | provider_cost_fen | INTEGER · NN · DEF |
| | total_cost_fen | INTEGER · NN |
| | cost_date | TEXT · NN |
| | source | TEXT · NN · DEF |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 期间核算与激励

### OPERATING_EXPENSES（运营费用）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | clinic_id → clinics | TEXT · NN · IDX |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| | category | TEXT · NN · IDX |
| | vendor_name | TEXT · NN · DEF |
| | amount_fen | INTEGER · NN |
| | incurred_on | TEXT · NN · IDX |
| | description_en | TEXT · NN |
| | description_zh | TEXT · NN |
| | status | TEXT · NN · DEF |
| FK | entered_by_user_id → app_users | TEXT · NULL |
| FK | approved_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### FINANCIAL_CLOSE_EXCEPTIONS（期间关闭异常）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NULL |
| | exception_type | TEXT · NN |
| | severity | TEXT · NN · IDX |
| | entity_type | TEXT · NN |
| | entity_id | TEXT · NN |
| | description_en | TEXT · NN |
| | description_zh | TEXT · NN |
| | status | TEXT · NN · IDX · DEF |
| | resolved_at | TEXT · NULL |
| | resolution_note | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### FINANCIAL_PERIOD_RESULTS（期间核算结果）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| FK | clinic_id → clinics | TEXT · NULL |
| | scope_key | TEXT · NN · IDX |
| | recognized_revenue_fen | INTEGER · NN |
| | treatment_cost_fen | INTEGER · NN |
| | operating_expense_fen | INTEGER · NN |
| | net_profit_fen | INTEGER · NN · IDX |
| | net_margin_basis_points | INTEGER · NN |
| | invoice_count | INTEGER · NN · DEF |
| | open_exception_count | INTEGER · NN · DEF |
| | calculation_inputs_json | TEXT · NN |
| | status | TEXT · NN · DEF |
| | calculated_at | TEXT · NN |
| | approved_at | TEXT · NULL |
| FK | approved_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

### INCENTIVE_POOLS（激励池）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | reporting_period_id → financial_reporting_periods | TEXT · NN · IDX |
| FK | network_result_id → financial_period_results | TEXT · NN |
| | rate_basis_points | INTEGER · NN · DEF |
| | eligible_net_profit_fen | INTEGER · NN |
| | pool_amount_fen | INTEGER · NN |
| | formula_evidence | TEXT · NN |
| | status | TEXT · NN · DEF |
| | approved_at | TEXT · NULL |
| FK | approved_by_user_id → app_users | TEXT · NULL |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表

## 扩张情景与候选诊所

### EXPANSION_SCENARIOS（扩张情景）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| | scenario_code | TEXT · NN · IDX |
| | version | INTEGER · NN · IDX · DEF |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | description_en | TEXT · NN |
| | description_zh | TEXT · NN |
| | forecast_horizon_months | INTEGER · NN · DEF |
| | status | TEXT · NN · IDX · DEF |
| | advisory_disclaimer_en | TEXT · NN |
| | advisory_disclaimer_zh | TEXT · NN |
| FK | created_by_user_id → app_users | TEXT · NULL |
| FK | approved_by_user_id → app_users | TEXT · NULL |
| | approved_at | TEXT · NULL |
| | created_at | TEXT · NN · DEF |
| | updated_at | TEXT · NN · IDX · DEF |

### EXPANSION_ASSUMPTIONS（扩张假设）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | scenario_id → expansion_scenarios | TEXT · NN · IDX |
| | assumption_key | TEXT · NN · IDX |
| | label_en | TEXT · NN |
| | label_zh | TEXT · NN |
| | value_text | TEXT · NN |
| | unit | TEXT · NN |
| | source_en | TEXT · NN |
| | source_zh | TEXT · NN |
| | created_at | TEXT · NN · DEF |

### EXPANSION_CANDIDATE_CLINICS（候选诊所）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | scenario_id → expansion_scenarios | TEXT · NN · IDX |
| | candidate_code | TEXT · NN · IDX |
| | name_en | TEXT · NN |
| | name_zh | TEXT · NN |
| | city_en | TEXT · NN |
| | city_zh | TEXT · NN |
| | district_en | TEXT · NN |
| | district_zh | TEXT · NN |
| | planned_chair_count | INTEGER · NN |
| | planned_dentist_count | INTEGER · NN |
| | planned_support_staff_count | INTEGER · NN |
| | forecast_quarterly_appointments | INTEGER · NN |
| | initial_capital_fen | INTEGER · NN |
| | quarterly_fixed_cost_fen | INTEGER · NN |
| | required_qualifications_json | TEXT · NN · DEF |
| | qualification_coverage_basis_points | INTEGER · NN |
| | travel_constraint_minutes | INTEGER · NN |
| | readiness_status | TEXT · NN · IDX · DEF |
| | created_at | TEXT · NN · DEF |

### EXPANSION_SCENARIO_RESULTS（扩张情景结果）

| | | |
|---|---|---|
| PK | id | TEXT · NN |
| FK | scenario_id → expansion_scenarios | TEXT · NN · IDX |
| | projected_quarter_revenue_fen | INTEGER · NN |
| | projected_treatment_cost_fen | INTEGER · NN |
| | projected_operating_expense_fen | INTEGER · NN |
| | projected_net_profit_fen | INTEGER · NN |
| | projected_margin_basis_points | INTEGER · NN |
| | payback_months | INTEGER · NN |
| | demand_coverage_basis_points | INTEGER · NN |
| | qualification_coverage_basis_points | INTEGER · NN |
| | composite_score | INTEGER · NN · IDX |
| | risk_level | TEXT · NN · IDX |
| | formula_evidence_json | TEXT · NN |
| | calculated_at | TEXT · NN |
| | created_at | TEXT · NN · DEF |

图例：PK 主键 | FK 外键 | NN 不可空 | NULL 可空 | IDX 索引 | DEF 默认值 | 箭头指向被引用父表